

Name:-Arpit Sunil Khodaskar
Roll No:-344481 (67) Class:- BE
Section :-A

Practical 1 :-Data VisualiZation Using Python.

#1

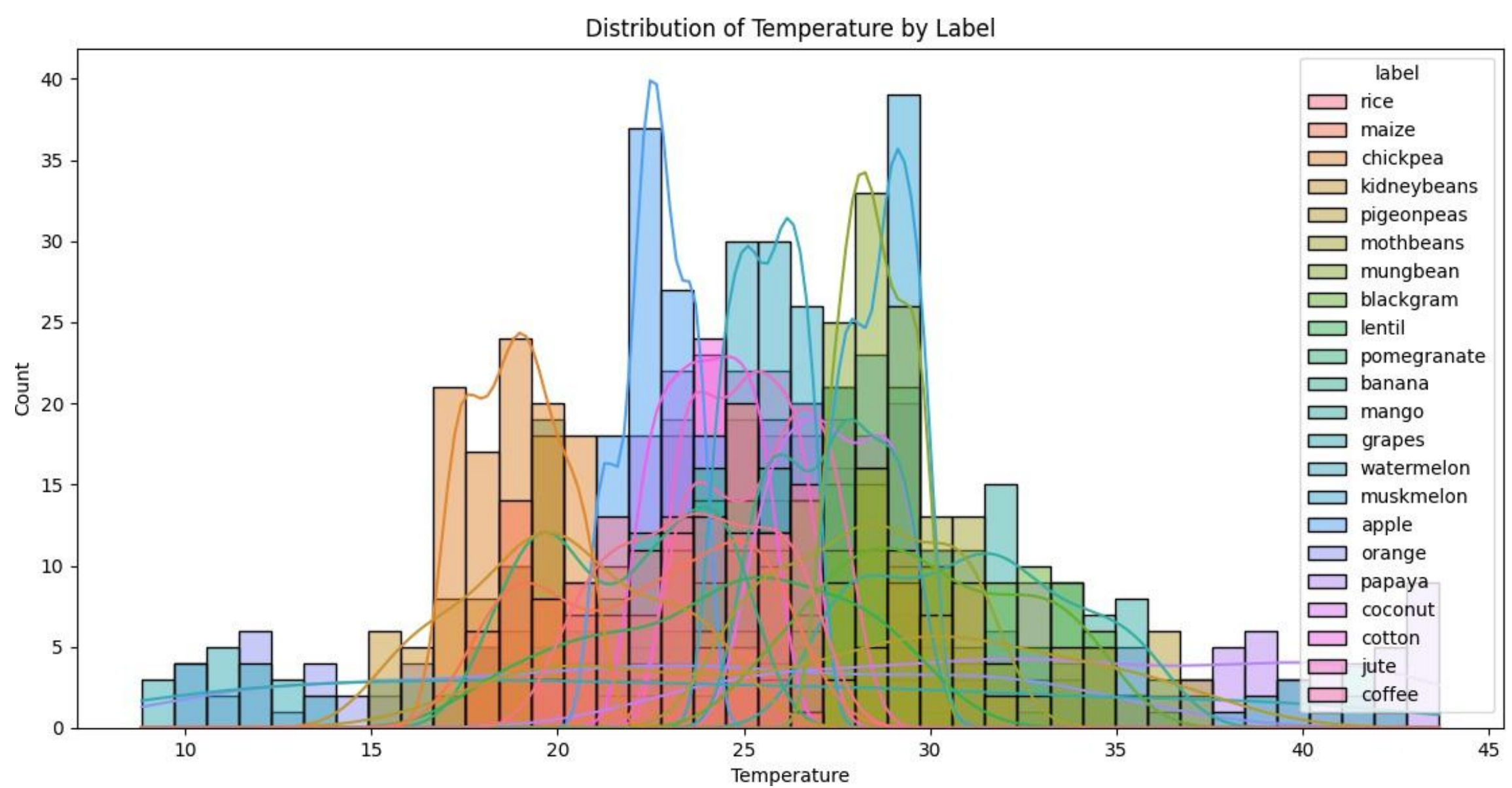
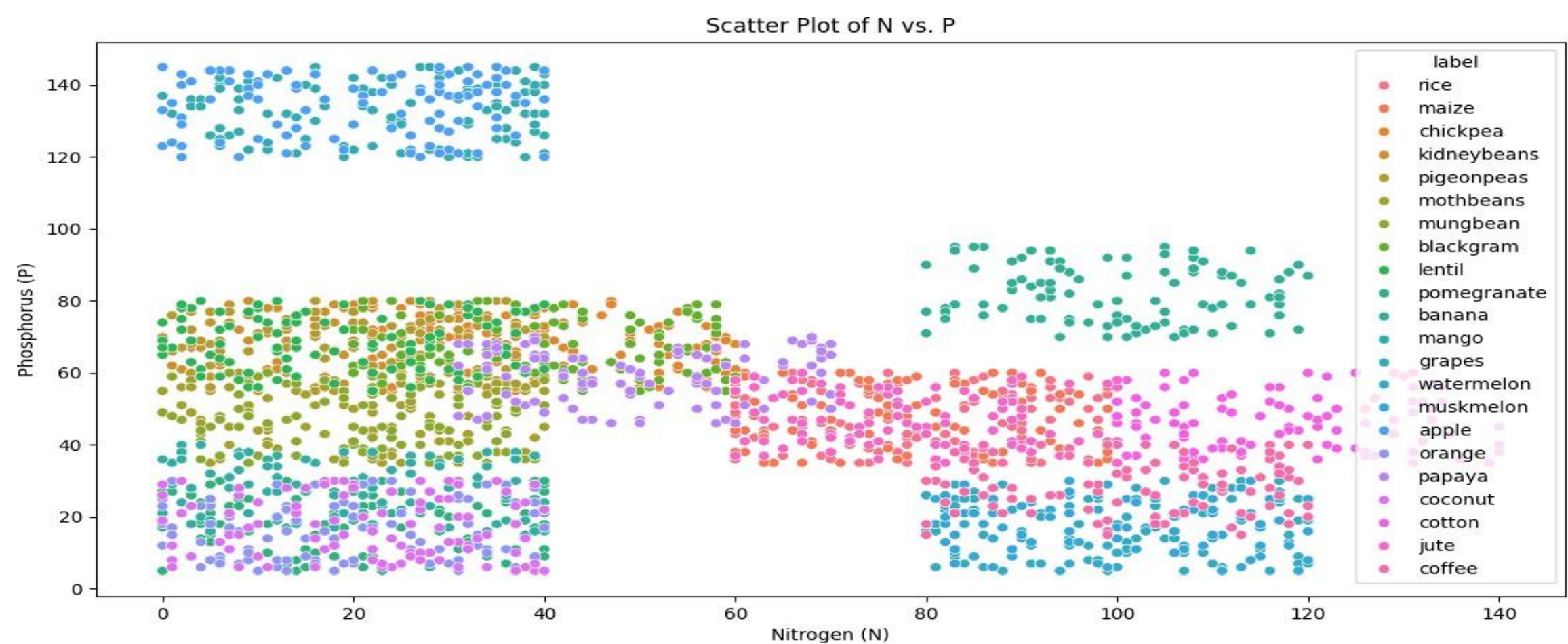
```
import pandas as pd
df = pd.read_csv('Crop_recommendation.csv')
print(df.head())
print(df.info())
print(df.describe())
#2#data visualization
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
#3#Example: Scatterplot of N vs. P colored by label
plt.figure(figsize=(10,6))
sns.scatterplot(x='N', y='P', hue='label', data=df)
plt.title('ScatterPlot of N vs. P')
plt.xlabel('Nitrogen(N)')
plt.ylabel('Phosphorus(P)')
plt.show()
#4#Example: Distribution of temperature by label
plt.figure(figsize=(10,6))
sns.histplot(data=df, x='temperature', hue='label', kde=True)
plt.title('Distribution of Temperature by Label')
plt.xlabel('Temperature')
plt.show()
#5#Example dataset data
data = {
    'N': [90, 85, 60, 74, 78],
    'P': [42, 58, 55, 35, 42],
    'label': ['A', 'B', 'A', 'B', 'A']
}
df = pd.DataFrame(data)
#Scatterplot of N vs. P colored by label
plt.figure(figsize=(8,6))
sns.scatterplot(x='N', y='P', hue='label', data=df, palette='Set1')
plt.title('ScatterPlot of N vs. P')
plt.xlabel('Nitrogen(N)')
plt.ylabel('Phosphorus(P)')
plt.show()
#6#Example dataset
data = {
    'Crop': ['Wheat', 'Rice', 'Maize', 'Barley'],
    'Yield': [4500, 3800, 5000, 4200]
}
df = pd.DataFrame(data)
```

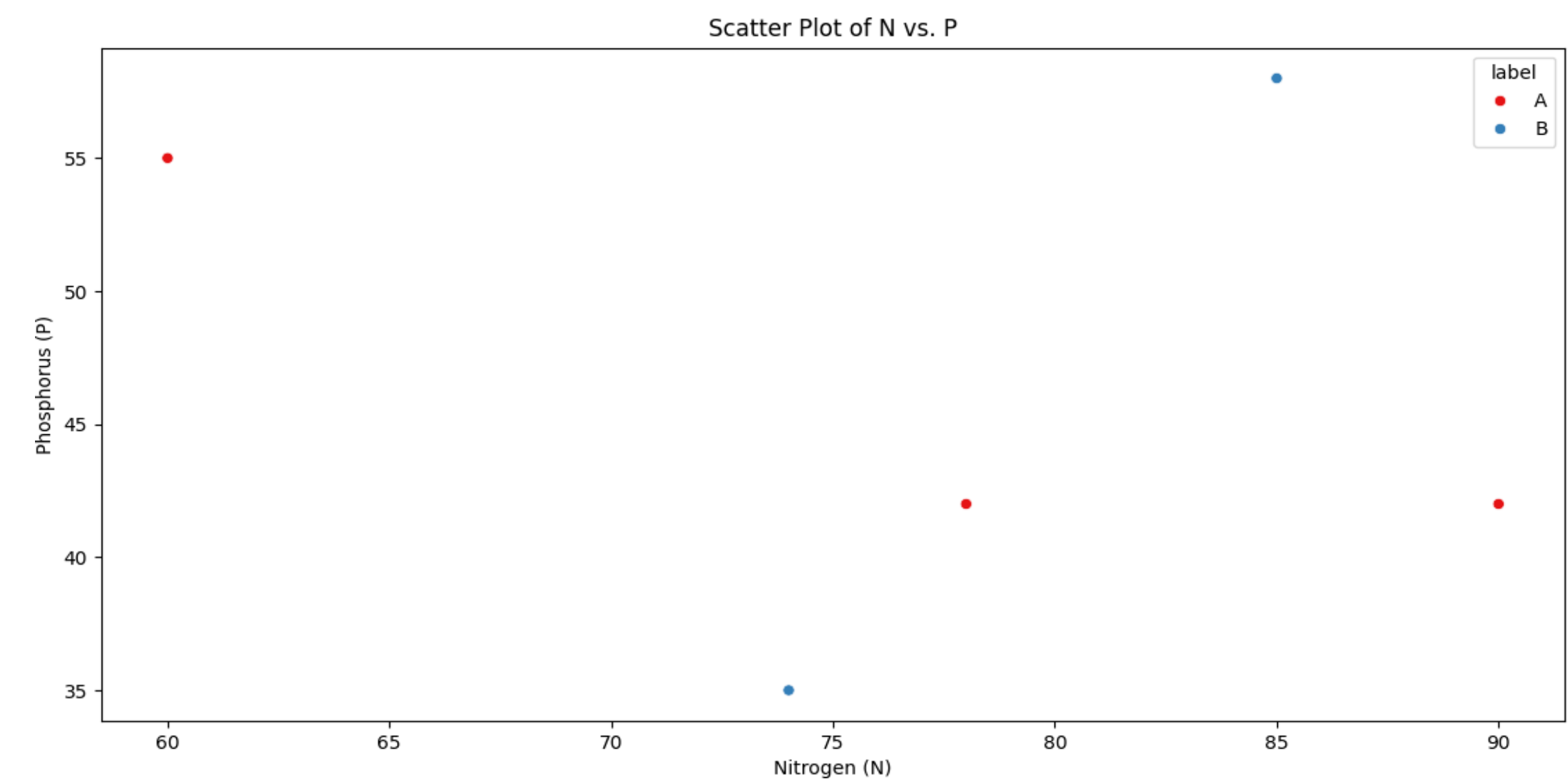
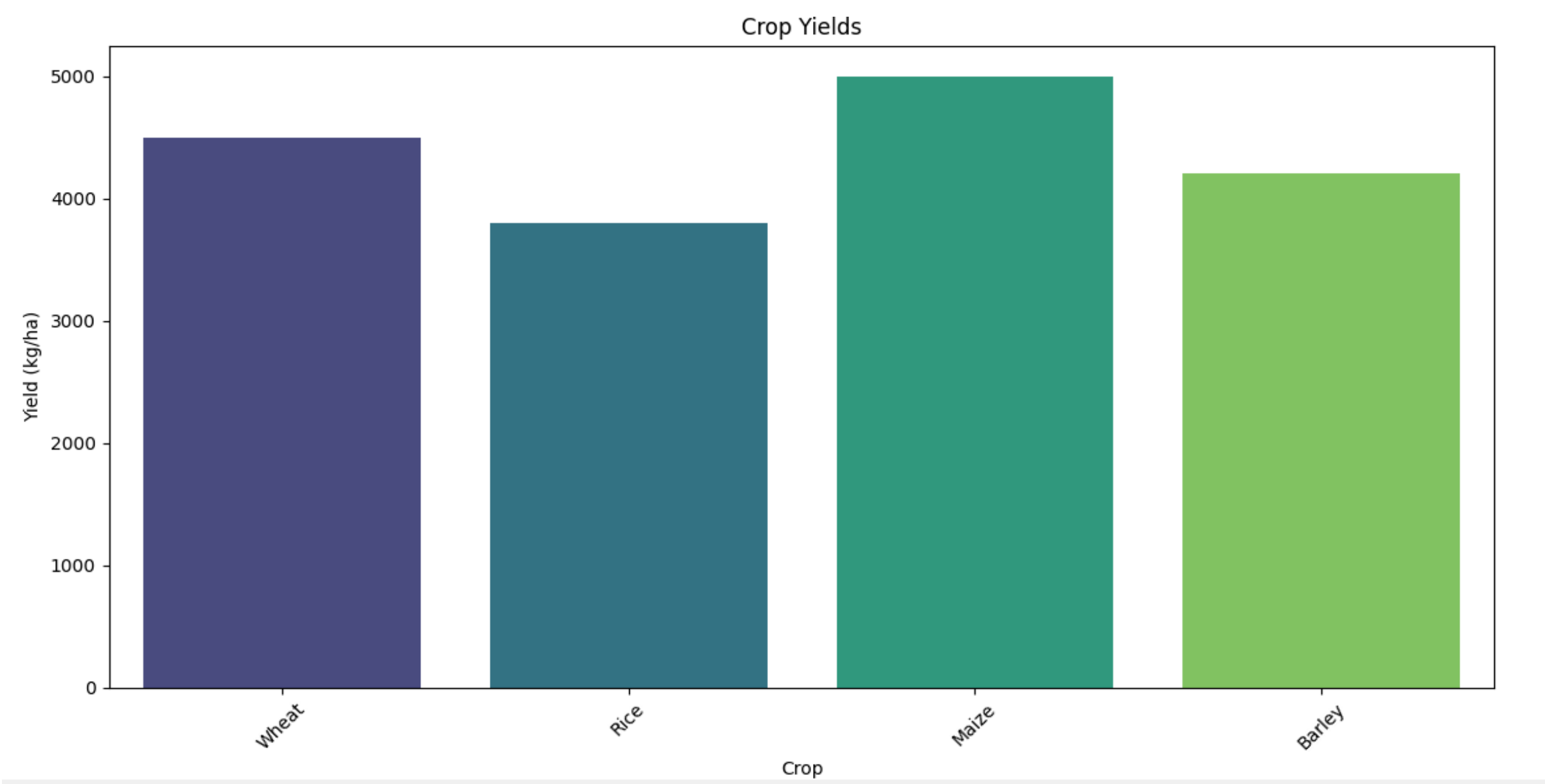
```

#Barchart of crop yields
plt.figure(figsize=(8, 6))
sns.barplot(x='Crop',y='Yield',data=df, palette='viridis')
plt.title('CropYields')
plt.xlabel('Crop')
plt.ylabel('Yield (kg/ha)')
plt.xticks(rotation=45)
plt.show()

```

OUTPUT:-





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Practical 2:-Implementation Of Django Stack.

index.html

```
<!DOCTYPEhtml>
<html>
<head>
<title>Local Library</title>
<linkhref="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
<style>
body{background:#f4f6f9;}
.sidebar{
height:100vh;
background:#1e293b;
padding-top:20px;
}
.sidebara{
display:block;
color:white; padding:12px20px;
text-decoration:none;
margin:5px;
border-radius:8px;
}
.sidebara:hover{
background:#3b82f6;
}
.content{
padding:40px;
}
.card-box{
border:none;
border-radius:20px;
box-shadow:010px25pxrgba(0,0,0,0.1);
}
</style>
</head>
<body>
<divclass="container-fluid">
<divclass="row">
<!--Sidebar-->
<divclass="col-md-2sidebar">
<h3 class="text-centertext-whiteb-4">📖 Library</h3>
<a href="/">🏠 Home</a>
```



```

    <linkhref="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"    rel="stylesheet">
</head>
<bodystyle="background:#f8f9fa;">
<navclass="navbarnavbar-darkbg-primary">
    <divclass="container">
        <a class="navbar-brand" href="/"><img alt="DjangoLibrary logo" data-bbox="408 168 428 188"/> DjangoLibrary</a>
    </div>
</nav>
<divclass="containermt-5">
    <divclass="card shadowwp-4rounded-4">
        <h2class="text-centertext-primarymb-4"><img alt="BooksCollection logo" data-bbox="458 253 478 273"/> BooksCollection</h2>
        <tableclass="tabletable-hover-table-bordered text-center">
            <trclass="table-dark">
                <th>BookTitle</th>
                <th>Author</th>
            </tr>
            {%forbookin books%}
            <tr>
                <td>{{book.title}}</td>
                <td>{{book.author}}</td>
            </tr>
            {%endfor%}
        </table>
        <divclass="text-centermt-3">
            <a href="/" class="btn btn-success">←BackHome</a>
        </div>
    </div>
</div>
</body>
</html>

```

authors.html

```

<!DOCTYPEhtml>
<html>
<head>
    <title>Authors</title>
    <linkhref="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"    rel="stylesheet">
</head>
<bodystyle="background:#eef2f7;">
<navclass="navbarnavbar-darkbg-success">
    <divclass="container">
        <a class="navbar-brand" href="/"><img alt="Authors logo" data-bbox="408 778 428 798"/> Authors</a>
    </div>
</nav>
<divclass="containermt-5">
<divclass="card shadowwp-4rounded-4">

```

```
<h2class="text-centertext-successmb-4">FamousAuthors</h2>
<ul class="list-group">
  {%fora in authors%}
  <li class="list-group-item">{{a}}</li>
  {%endfor%}
</ul>
<divclass="text-centermt-4">
  <a href="/" class="btn btn-primary">←BackHome</a>
</div>
</div>
</div>
</body>
</html>
```

borrowed.html

```
<!DOCTYPEhtml>
<html>
<head>
<title>Borrowed Books</title>
<linkhref="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<bodystyle="background:#eef2f7;">
<divclass="containermt-5">
<divclass="card shadow-lg rounded-4 border-0 p-4">
<h1 class="text-center text-warning">📖 MyBorrowed Books</h1 >
<tableclass="table table-bordered table-hover mt-4">
<trclass="table-dark">
<th>BookName</th>
<th>Status</th>
<th>DueDate</th>
</tr>
<tr>
<td>PythonBasics</td>
<td>Issued</td>
<td>1 0May2026</td>
</tr>
<tr>
<td>DjangoGuide</td>
<td>Issued</td>
<td>1 5May2026</td>
</tr>
</table>
<divclass="text-center mt-3">
<a href="/" class="btn btn-primary">←BackHome</a>
</div>
</div>
</div>
```



```
</body>
</html>
```

admin.py

```
from django.contrib import admin
from .models import Book
admin.site.register(Book)
```

apps.py

```
from django.apps import AppConfig
class CatalogConfig(AppConfig):
    name = 'catalog'
```

models.py

```
from django.db import models
class Book(models.Model):
    title = models.CharField(max_length=100)
    author = models.CharField(max_length=100)

    def __str__(self):
        return self.title
```

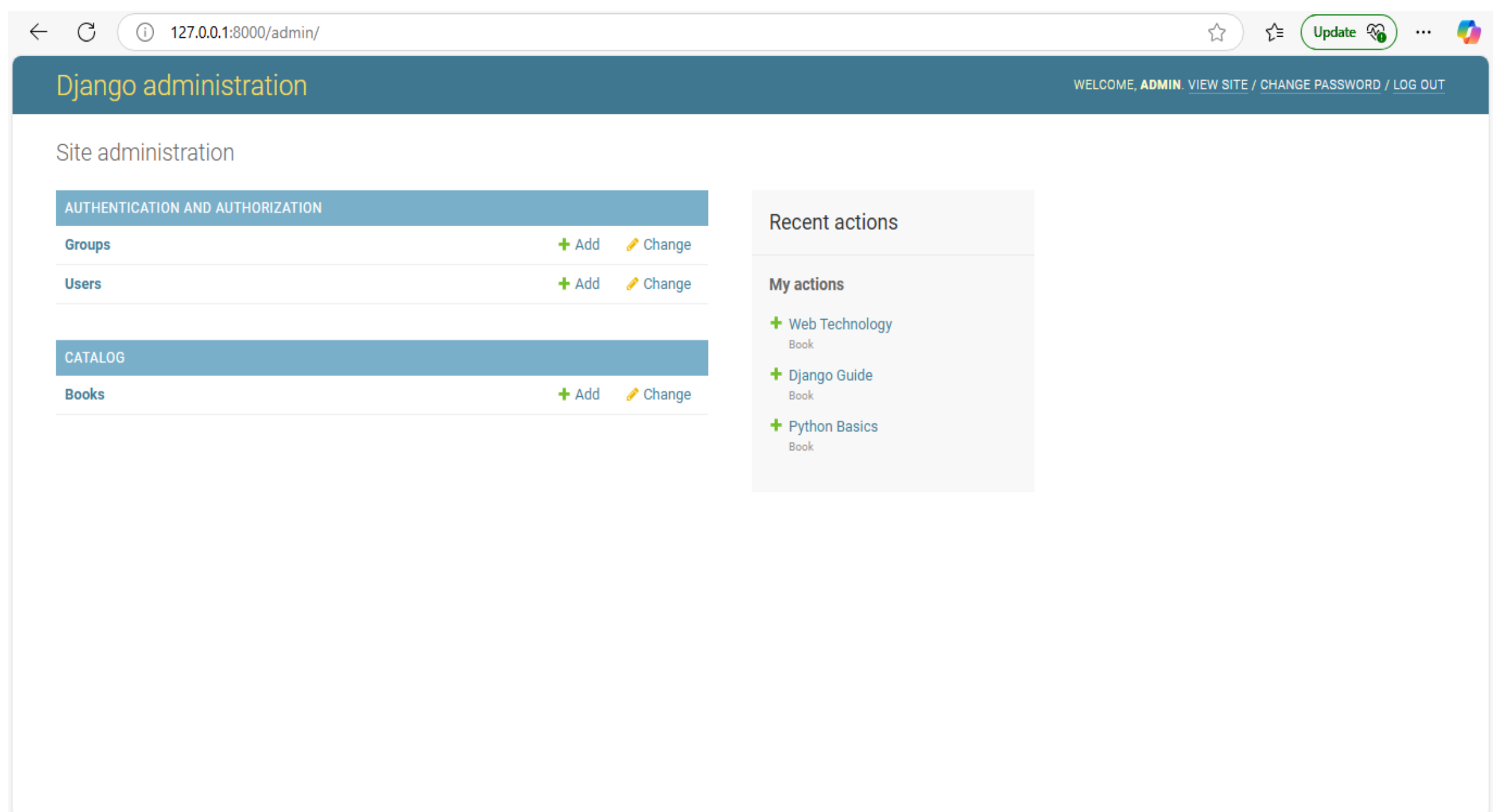
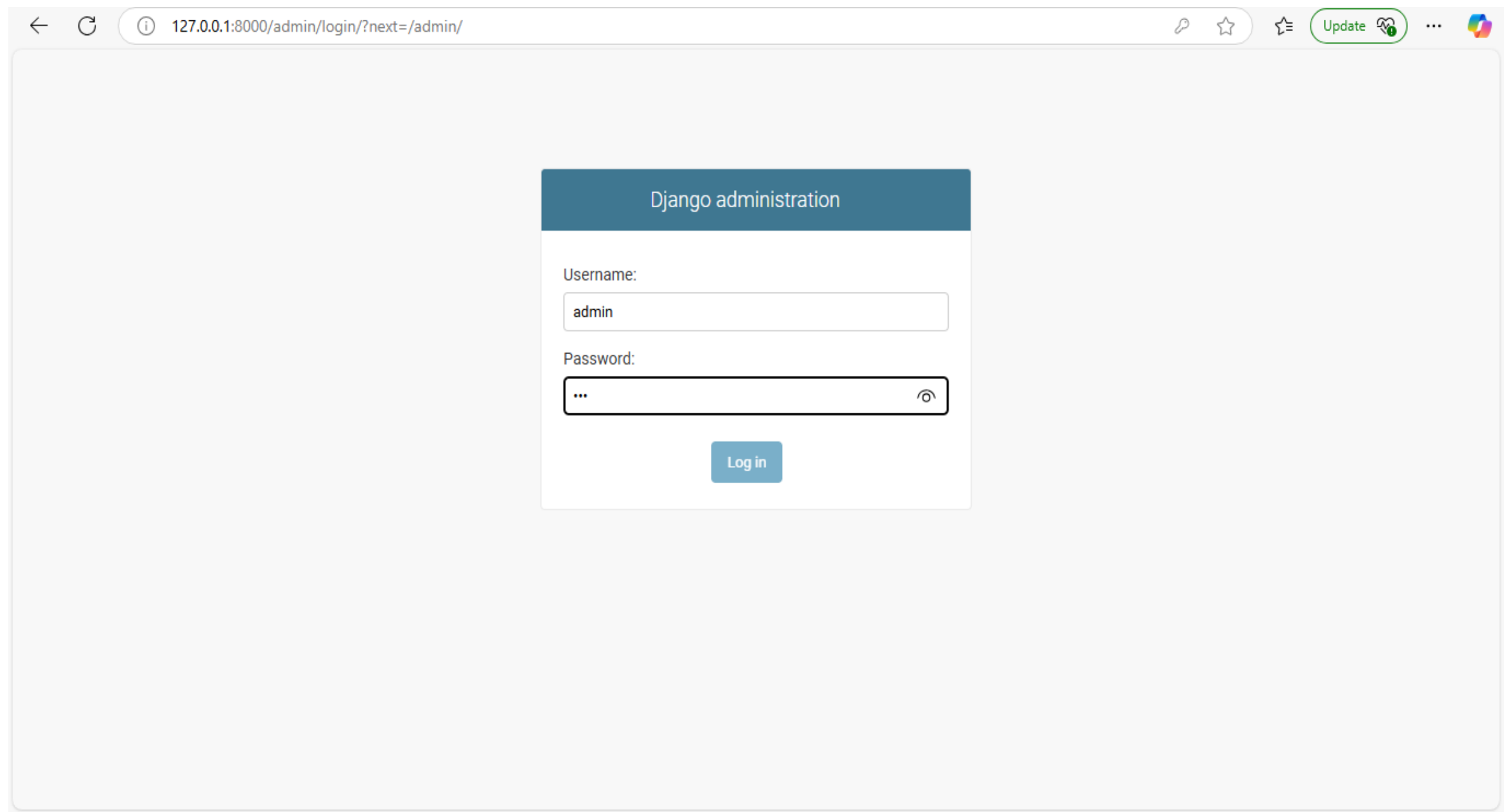
urls.py

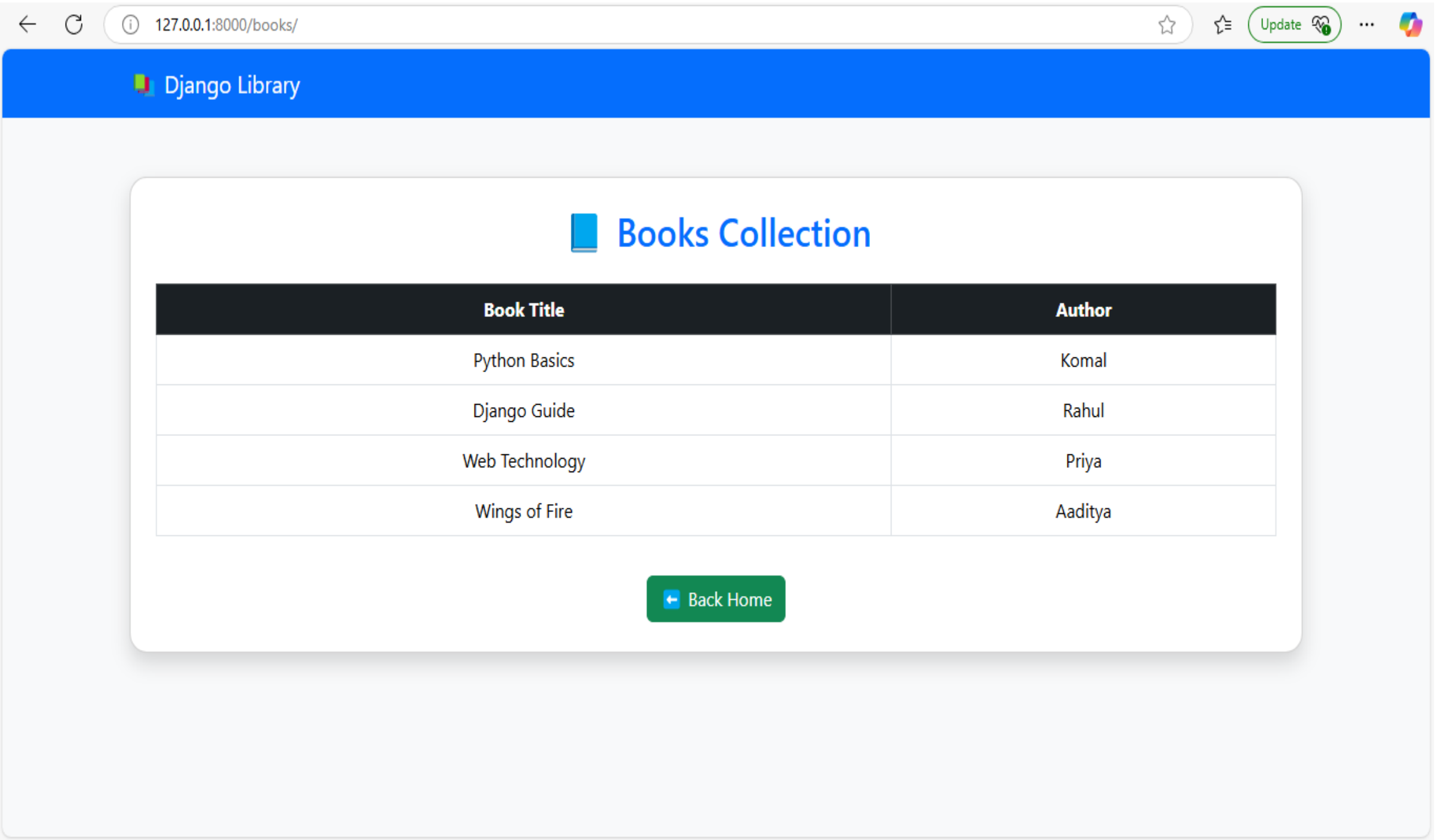
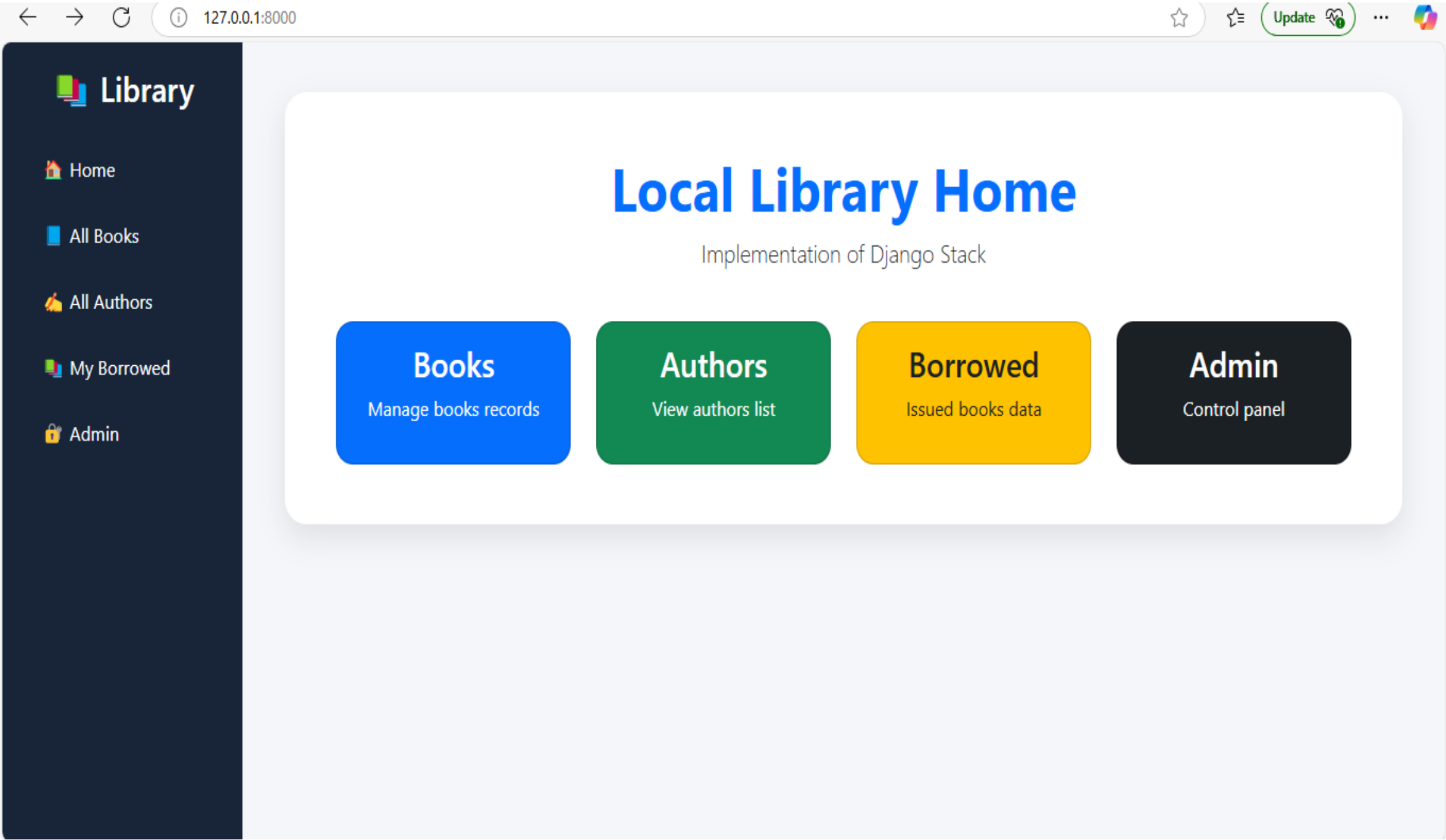
```
from django.urls import path
from . import views
urlpatterns = [
    path('', views.home, name='home'),
    path('books/', views.books, name='books'),
    path('authors/', views.authors, name='authors'),
    path('borrowed/', views.borrowed, name='borrowed'),
]
```

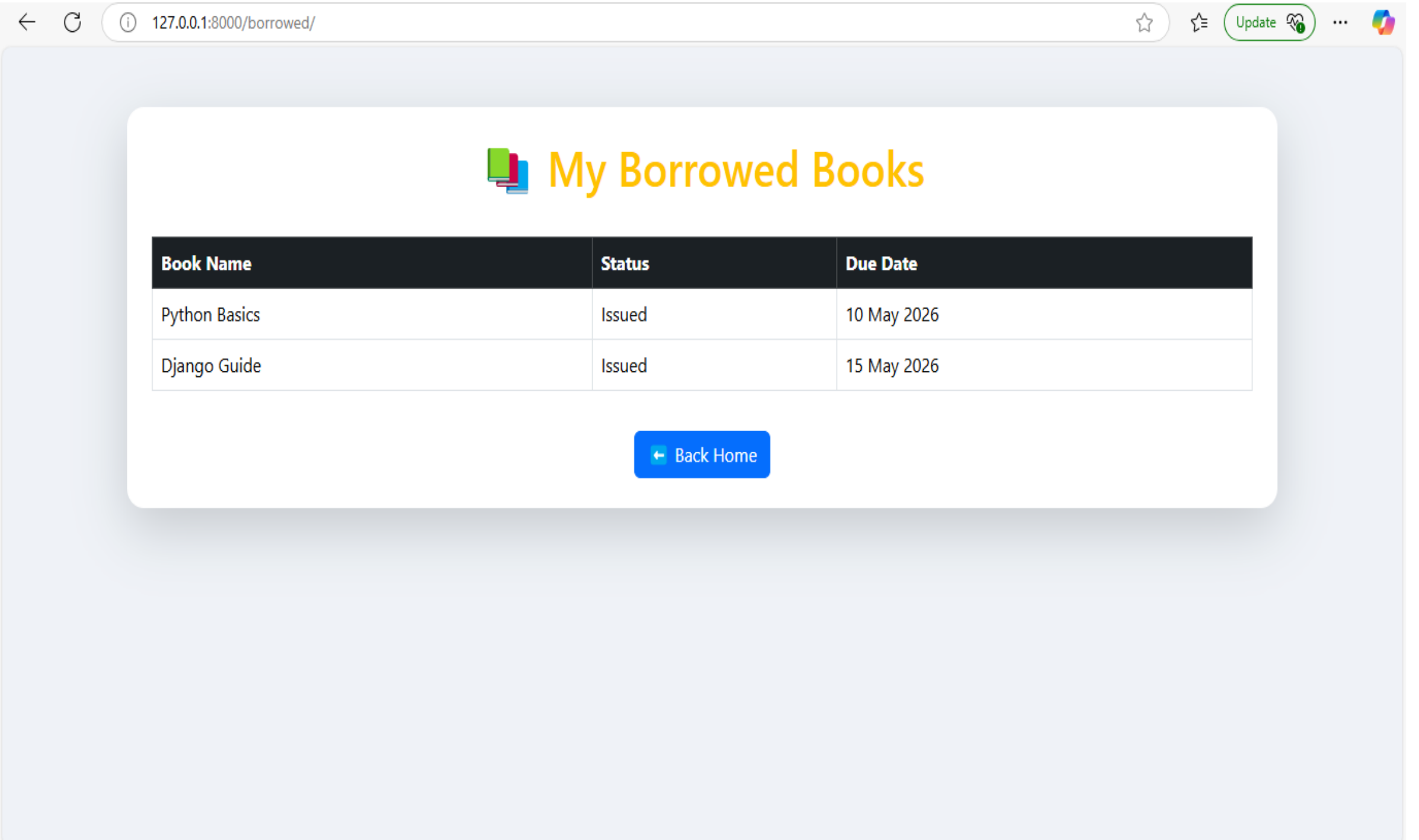
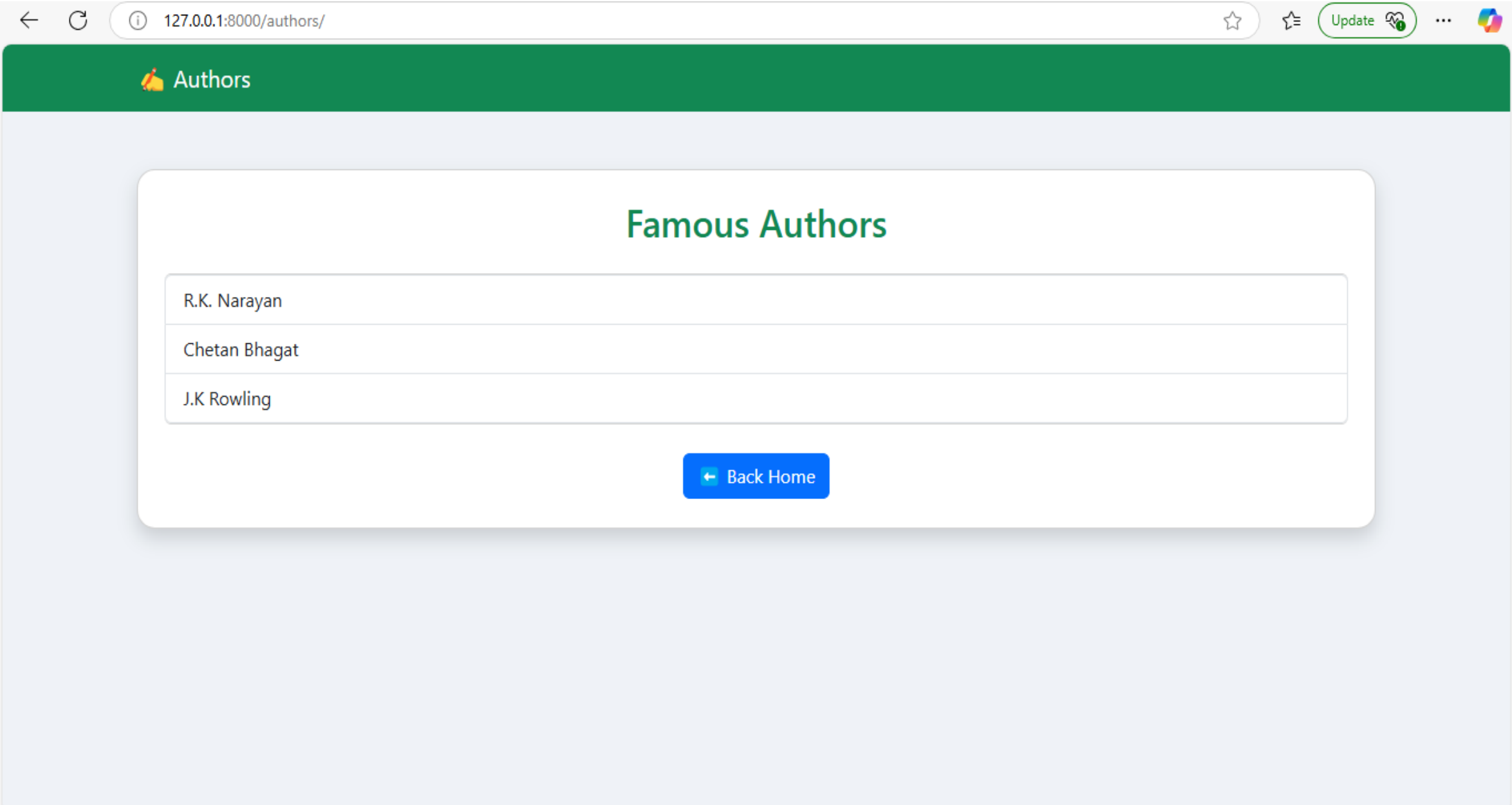
views.py

```
from django.shortcuts import render
from .models import Book
def home(request):
    return render(request, 'index.html')
def books(request):
    all_books = Book.objects.all()
    return render(request, 'books.html', {'books': all_books})
def authors(request):
    data = {'authors': ['R.K. Narayan', 'Chetan Bhagat', 'J.K. Rowling']}
    return render(request, 'authors.html', data)
def borrowed(request):
    return render(request, 'borrowed.html')
```

OUTPUT:-







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Practical 3 :-Createa Rubyon Railsanapplication.

todo-list/app/models/application_record.rb

```
classApplicationRecord<ActiveRecord::Base primary_abstract_class
end
```

new.html.erb

```
<divclass="topbar">
Newtask
<%=link_to"Goback", tasks_path, class:"back-btn"%>
</div>
```

```
<%=render"form", task:@task%>
```

show.html.erb

```
<divclass="header">TaskDetails</div>
<pclass="notice"><%= notice%></p>
<divclass="form-box">
<divclass="task-box">
<b>Title:</b><%= @task.title%>
</div>
<divclass="task-box">
<b>Status:</b>
<%=@task.completed?" Completed" : " Pending"%>
</div>
<divclass="center">
<%=link_to"Edit",edit_task_path(@task),class:"btn"%>
<%=link_to"Back",tasks_path,class:"btn"%>
<%=link_to"Delete", @task, method: :delete, data:
{confirm: "Areyousure?"}, class:"btn" %>
</div>
</div>
```

index.html.erb

```
<divclass="header">iTasks</div>
<pclass="notice"><%= notice%></p>
<divclass="container">
<h2>To-Do</h2>
<%=@tasks.where(completed:false).eachdo|task| %>
<divclass="task-box">
```

```

<span><%= task.title%></span>
<span>
<%=link_to"✎",edit_task_path(task) %>
<%=button_to"🗑️",task,
method: :delete,
form: {style: "display:inline;"},
data: {turbo_confirm: "Delete?"},
class: "delete-btn" %>
</span>
</div>
<%end%>
<divclass="center">
<%=link_to"Newtask", new_task_path,class:"btn"%>
</div>
<h2>Completed</h2>
<%=@tasks.where(completed:true).each do |task| %>
<divclass="task-box">
<span><%= task.title%></span>
<span>
<%=button_to"🗑️",task,
method: :delete,
form: {style: "display:inline;"},
data: {turbo_confirm: "Delete?"},
class: "delete-btn" %>
</span>
</div>
<%end%>
</div>

```

task.html.erb

```

<divid="<%= dom_id task%>">
  <p>
    <strong>Title:</strong>
    <%=task.title%>
  </p>
  <p>
    <strong>Completed:</strong>
    <%=task.completed %>
  </p>
</div>

```

form.html.erb

```

<divclass="main-form">

<%=form_with(model: task) do |form| %>
<%iftask.errors.any?%>

```



```

<divclass="error-box">
<h3><%= pluralize(task.errors.count,"error") %>prohibited thistaskfrombeing saved:</h3>
<ul>
<%task.errors.full_messages.eachdo|msg| %>
<li><%=msg %></li>
<%end %>
</ul>
</div>
<%end %>
<divclass="row">
<label>Task</label>
<%=form.text_field :title%>
</div>
<divclass="row">
<label>Details</label>
<%=form.text_area:details%>
</div>
<divclass="row">
<label>Duedate</label>
<%=form.date_field :due_date %>
</div>
<divclass="row">
<label>Completed</label>
<%=form.check_box:completed, class:"checkbox" %>
</div>
<divclass="center">
<%=form.submit class:"submit-btn" %>
</div>
<%end %>
</div>

```

Application.html.erb

```

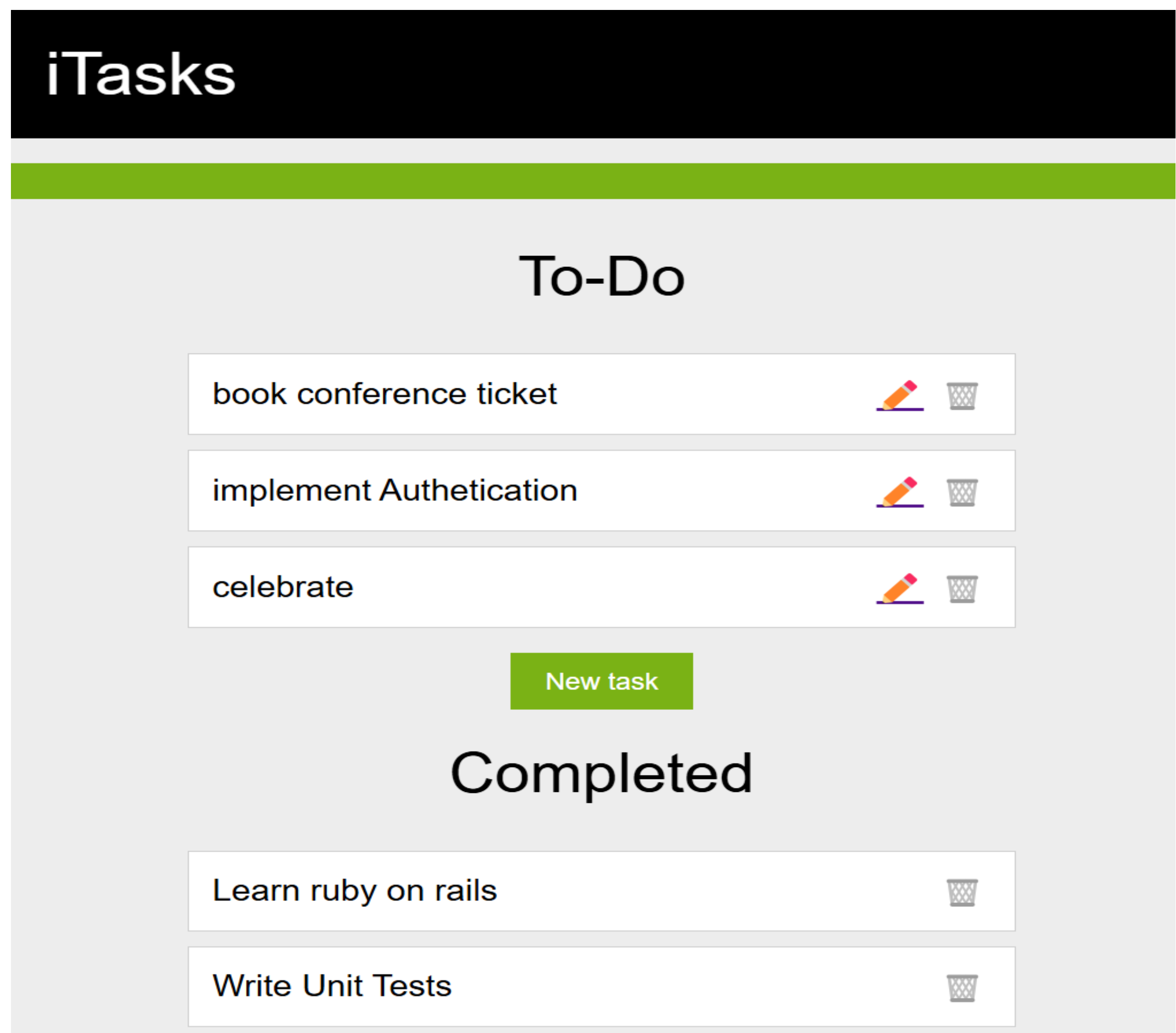
<!DOCTYPEhtml>
<html>
<head>
<title>Todoapp</title>
<meta name="viewport" content="width=device-width,initial-scale=1 ">
<%=csrf_meta_tags%>
<%=csp_meta_tag %>
<%=stylesheet_link_tag "application","data-turbo-track": "reload" %>
</head>
<body>
<%=yield %>
</body>
</html>

```

Mailer.html.erb

```
<!DOCTYPEhtml>
<html>
  <head>
    <metahttp-equiv="Content-Type" content="text/html; charset=utf-8">
    <style>
      /*Email stylesneedtobeinline*/
    </style>
  </head>
  <body>
    <%=yield %>
  </body>
</html>
```

OUTPUT:-



New task

[Go back](#)

Task

Details

Due
date

dd-mm-yyyy



Completed ☐

Create Task